St. Philomena's College (Autonomous), Mysore
PG Department of Chemistry
Question Bank (Revised Curriculum 2018 onwards)
Second Year- Third Semester (2019-2021 Batch)
Course Title (Paper Title):
Chemistry of Natural Products QP Code:55203

Unit	SI. No	Questions	Marks
1	1.	What are steroids? Mention the structural unit present in steroids.	**2**
1	2.	Write the IUPAC name and stereochemistry of cholesterol.	**2**
1	3.	How to determine the angular methyl group at C-10 position of cholesterol?	**2**
1	4.	Write the structure of cholesterol with its function.	**2**
1	5.	What are ergosterol? Write any two biological significance.	**2**
1	6.	Write the irradiation product of ergosterol	**2**
1	7.	Write the nomenclature and structure of ergosterol	**2**
1	8.	Write the IUPAC name and structure of ergosterol	**2**
1	9.	Write the stereochemistry of ergosterol.	**2**
1	10.	Give any four biological application of ergosterol	**2**
1	11.	Give the biological importance of bile acids.	**2**
1	12.	Write biological applications of bile acids	**2**
1	13.	Write the IUPAC name and structure of bile acids.	**2**
1	14.	What are bile acids? Write any two biological significance	**2**
1	15.	Give any two applications of estrone.	**2**
1	16.	Write the IUPAC name and structure of estrone.	**2**

1	17.	What is estrone? Write any two biological significance	**2**
1	18.	Write the stereochemistry of estrone.	**2**
1	19.	What is IUPAC name of progesterone	**2**
1	20.	Write the stereochemistry of progesterone	**2**
1	21.	Mention the biological importance of progesterone.	**2**
1	22.	Write the IUPAC name and structure of progesterone.	**2**
1	23.	What is progesterone? Write any two biological significance	**2**
1	24.	Write the nomenclature and structure of testosterone.	**2**
1	25.	Write the IUPAC name and structure of testosterone.	**2**
1	26.	Write the stereochemistry of testosterone	**2**
1	27.	What is testosterone? Write any two biological significance	**2**
1	28.	What are androsterone? Write any two biological significance.	**2**
1	29.	Mention the importance of androsterone.	**2**
1	30.	Write the IUPAC name and structure of androsterone.	**2**
1	31.	Write the stereochemistry of androsterone.	**2**
1	32.	What are corticosteroids? Write any two biological significance	**2**
1	33.	Write the structure of carticosterone	**2**
1	34.	Write the stereochemistry of carticosterone	**2**
1	35.	What are carticosterone? Write any two biological significance.	**2**
1	36.	Write the IUPAC name and structure carticosterone	**2**
2	37.	What are alkaloids? Give their occurrence.	**2**

2	38.	Define alkaloids. Write their occurrence	**2**
2	39.	Give the chemical methods of classification of alkaloids.	**2**
2	40.	Give a general method for isolation of Alkaloids.	**2**
2	41.	Write a general method for isolation of Alkaloids.	**2**
2	42.	Explain any one method for isolation of Alkaloids.	**2**
2	43.	How do you prove the presence of methoxyl group in general methods for determining structures of alkaloids?	**2**
2	44.	How do you prove the presence of carbonyl group in general methods for determining structures of alkaloids?	**2**
2	45.	How do you prove the presence of carboxylic group in general methods for determining structures of alkaloids?	**2**
2	46.	How do you prove the presence of ester group in general methods for determining structures of alkaloids?	**2**
2	47.	Write the chemical methods of classification of alkaloids.	**2**
2	48.	Give the biological importance of alkaloids.	**2**
2	49.	Mention the biological importance of alkaloids.	**2**
2	50.	Listout the biological importance of alkaloids.	**2**
2	51.	Outline the biological importance of alkaloids.	**2**
2	52.	Give the general characteristic of alkaloids.	**2**
2	53.	How do you prove the presence of an olefinic linkage in Quinine alkaloids?	**2**
2	54.	Prove the presence of an olefinic linkage in Quinine alkaloids.	**2**
2	55.	Explain the presence of an olefinic linkage in Quinine alkaloids.	**2**
2	56.	Describe the presence of an olefinic linkage in Quinine alkaloids.	**2**
2	57.	How do you prove the presence of quininic acid and meroquinene in Quinine alkaloids?	**2**
2	58.	Prove the presence of quininic acid and meroquinene in Quinine alkaloids?	**2**
2	59.	Explain the presence of quininic acid and meroquinene in Quinine	**2**

		alkaloids?	
2	60.	Describe the presence of quininic acid and meroquinene in Quinine alkaloids?	**2**
2	61.	How do you prove the presence of methoxyl group in Quinine alkaloids?	**2**
2	62.	Prove the presence of methoxyl group in Quinine alkaloids?	**2**
2	63.	Explain the presence of methoxyl group in Quinine alkaloids?	**2**
2	64.	Describe the presence of methoxyl group in Quinine alkaloids?	**2**
2	65.	Write the nature of oxygen atoms in morphine alkaloids?	**2**
2	66.	Give the applications of lysergic acid.	**2**
2	67.	Write any two applications of lysergic acid.	**2**
2	68.	List out any two applications of lysergic acid.	**2**
2	69.	Mention any two applications of lysergic acid.	**2**
2	70.	How do you prove the presence of pyridine nucleus in pyridine alkaloids?	**2**
2	71.	Mention the biological importance of Morphine.	**2**
2	72.	Give the biological importance of Morphine.	**2**
2	73.	List out the boilogical importance of Morphine.	**2**
3	74.	Write the sources of vitamins. Mention its importance.	**2**
3	75.	Why vitamins are called accessary dietary factors.	**2**
3	76.	What are the diseases caused due to deficiency of vitamin-A	**2**
3	77.	What are the diseases caused due to deficiency of vitamin-B	**2**
3	78.	What are the diseases caused due to deficiency of vitamin-B3	**2**
3	79.	What are the diseases caused due to deficiency of vitamin-B_c	**2**
3	80.	What are the diseases caused due to deficiency of vitamin-B_5	**2**
3	81.	What are the diseases disease caused due to deficiency of vitamin-C	**2**
3	82.	What are the diseases caused due to deficiency of vitamin-A	**2**
3	83.	What are the diseases caused due to deficiency of vitamin-D	**2**

3	84.	What are the diseases caused due to deficiency of vitamin-E	**2**
3	85.	What are the diseases caused due to deficiency of vitamin-K	**2**
3	86.	Write the classification of vitamins	**2**
3	87.	Write any two sources of vitamin-A	**2**
3	88.	Write any two sources of vitamin-B_1	**2**
3	89.	Write any two sources of vitamin-C	**2**
3	90.	Write any two sources of vitamin-D_3	**2**
3	91.	Write any two sources of vitamin-E	**2**
3	92.	Write any two sources of vitamin-K	**2**
3	93.	What are precursors give example	**2**
3	94.	What are provitamins give example	**2**
3	95.	Write the structure of erosterol	**2**
3	96.	Write the structure of Beta carotene	**2**
3	97.	Give an example for fat soluble vitamins	**2**
3	98.	Give examples for water soluble vitamins	**2**
3	99.	What is the name adapted for vitamin-A_1 and A_2	**2**
3	100.	What is the name adapted for vitamin-B_1 and B_2	**2**
3	101.	What is the name adapted for vitamin-B_6 and B_{12}	**2**
3	102.	What is the name adapted for vitamin-C and D_2	**2**
3	103.	What is the name adapted for vitamin-D_3 and E	**2**
3	104.	What is the name adapted for vitamin-H and D_2	**2**
3	105.	Write the structure of ergo calciferol	**2**
3	106.	Write the structure of retinol	**2**
3	107.	Write the structure of 3,4 dehydroretinol	**2**
3	108.	Write the biological significance of thiamine	**2**

3	109.	What is the action of vit-B1purophosphate	**2**
3	110.	Write the IUPAC name for Pyridoxine	**2**
3	111.	Write the structures of pyridoxine and pyridoxal	**2**
3	112.	Write the structures of pyridoxal and pyridoxamine	**2**
3	113.	How do you prove vitamin-A1 contains beta-ionone nucleus	**2**
3	114.	How egg white injury occurs due to deficiency of biotin.	**2**
3	115.	Write the structure of vitamin-H	**2**
3	116.	Write the synthesis of valeric acid	**2**
3	117.	Write the structure of 3,4 dehydroretinol	**2**
3	118.	Write the biological significance of thiamine	**2**
3	119.	What is the action of vit-B1purophosphate	**2**
3	120.	Write the IUPAC name for pyridoxine	**2**
3	121.	How do you prove vitamin-A_1 contains beta-ionone nucleus	**2**
3	122.	How egg white injury occurs due to deficiency of biotin.	**2**
3	123.	Write the structure of vitamin-H	**2**
3	124.	Write the synthesis of valeric acid	**2**
3	125.	Write the biological significance of thiamine	**2**
3	126.	Vitamin-C is known as hexuronic acid. Justify	**2**
3	127.	What are vitamins? How they are classified?	**2**
3	128.	Mention the sources and importance of vitamin-K	**2**
3	129.	What happens when thiamine is treated with sodium sulphite? Give the reaction.	**2**
3	130.	Write the structure of ascorbic acid and mention the stereocentres.	**2**
2	131.	Write a note on classification of alkaloids.	4
2	132.	Give the classification of alkaloids.	4

2	133.	Outline the classification of alkaloids.	4
2	134.	Write the classification of alkaloids.	4
2	135.	Elaborate the classification of alkaloids.	4
2	136.	Explain the isolation of alkaloids.	4
2	137.	Describe the isolation of alkaloids.	4
2	138.	Outline the isolation of alkaloids.	4
2	139.	Elaborate the isolation of alkaloids.	4
2	140.	Give the synthesis of lysergic acid.	4
2	141.	Explain the synthesis of lysergic acid.	4
2	142.	Describe the synthesis of lysergic acid.	4
2	143.	Outline the synthesis of lysergic acid.	4
2	144.	Write a note on the synthesis of lysergic acid.	4
2	145.	Give the synthesis of nicotine.	4
2	146.	Explain the synthesis of nicotine.	4
2	147.	Describe the synthesis of nicotine.	4
2	148.	Outline the synthesis of nicotine.	4
1	149.	Outline the synthesis of cholesterol.	5
1	150.	Describe the complete synthesis of cholesterol.	5
1	151.	Elucidate the structure of cholesterol.	5
1	152.	Summarize a report on the elucidation of cholesterol.	5
1	153.	Write a note the determination of angular methyl group at C-17, functional group (-OH) and double bond in cholesterol.	5
1	154.	Explain the determination of functional group (-OH) and double bond in cholesterol.	5
1	155.	Describe the importance of and its irradiation product.	5

1	156.	Summarize a report on irradiation product of ergosterol	5
1	157.	Write a note on ergosterol and its irradiation product	5
1	158.	What is ergosterol? Mention the biological significance of ergosterol	5
1	159.	Write a note on biological importance of bile acids.	5
1	160.	Explain the structure of bile acids, indicate the stereochemistry and hence also write its biological significance.	5
1	161.	What are bile acids? Write a note on its biological importance	5
1	162.	Write the structure and biological significance of bile acids	5
1	163.	Explain the biological significance of estrone	5
1	164.	What is estrone? Hence, write a brief note on its biological importance.	5
1	165.	Explain the structure of estrone, indicate the stereochemistry and hence also write its biological significance	5
1	166.	Summarize a report on the biological importance of estrone	5
1	167.	Explain the biological significance of progesterone	5
1	168.	Summarize a report on the biological importance of progesterone	5
1	169.	What is progesterone? Hence, write a brief note on its biological importance.	5
1	170.	Explain the structure of progesterone, indicate the stereochemistry and hence also write its biological significance	5
1	171.	Write a note on biological importance of testosterone.	5
1	172.	Describe stereochemistry, nomenclature and biological importance of testosterone.	5
1	173.	Explain the structure of testosterone, indicate the stereochemistry and hence also write its biological significance.	5
1	174.	What is testosterone? Mention its biological significance in both male and female.	5
1	175.	Write a note on biological importance of androsterone.	5

1	176.	What are androsterone. Write a brief report on its biological significance	5
1	177.	Describe stereochemistry, nomenclature and biological importance of androsterone.	5
1	178.	Explain the structure of androsterone, indicate the stereochemistry and hence also write its biological significance.	5
1	179.	Write the biological significance of corticosteroids.	5
1	180.	Describe the structure of corticosterone, indicate the stereochemistry and hence also write its biological significance.	5
1	181.	What is corticosterone? Hence, write a brief note on its biological importance.	5
2	182.	Give the occurrence and classification of alkaloids.	5
2	183.	Define alkaloids and outline their classification.	5
2	184.	Write the occurrence and classification of alkaloids.	5
2	185.	What are alkaloids? Elaborate the classification of alkaloids.	5
2	186.	Define alkaloids. Explain the isolation of alkaloids.	5
2	187.	Give the occurrence of alkaloids and describe their isolation.	5
2	188.	Outline the occurrence and isolation of alkaloids.	5
2	189.	Elaborate the occurrence and isolation of alkaloids.	5
2	190.	Explain the general methods for determining the structures of alkaloids.	5
2	191.	Describe the methods for determining the structures of alkaloids.	5
2	192.	Outline the general methods for determining the structures of alkaloids.	5
2	193.	Elaborate the methods for determining the structures of alkaloids.	5
2	194.	Write a note on the general methods for determining the structures of alkaloids.	5
2	195.	How do you elucidate the structure of quinine?	5
2	196.	Elucidate the structure of quinine.	5
2	197.	Explain the elucidation of the structure of quinine.	5

2	198.	Outline the structure of quinine.	5
2	199.	Elaborate the structure of quinine.	5
2	200.	Give the synthesis of quinine.	5
2	201.	Explain the synthesis of quinine.	5
2	202.	Describe the synthesis of quinine.	5
2	203.	Outline the synthesis of quinine.	5
2	204.	Write a note on the synthesis of quinine.	5
2	205.	How do you elucidate the structure of morphine?	5
2	206.	Elucidate the structure of morphine.	5
2	207.	Explain the elucidation of the structure of morphine.	5
2	208.	Outline the structure of morphine.	5
2	209.	Elaborate the structure of morphine.	5
2	210.	Give the synthesis of morphine.	5
2	211.	Explain the synthesis of morphine.	5
2	212.	Describe the synthesis of morphine.	5
2	213.	Outline the synthesis of morphine.	5
2	214.	Write a note on the synthesis of morphine.	5
2	215.	How do you elucidate the structure of reserpine?	5
2	216.	Elucidate the structure of reserpine.	5
2	217.	Explain the elucidation of the structure of reserpine.	5
2	218.	Outline the structure of reserpine.	5
2	219.	Elaborate the structure of reserpine.	5
2	220.	Give the synthesis of reserpine.	5
2	221.	Explain the synthesis of reserpine.	5
2	222.	Describe the synthesis of reserpine.	5

2	223.	Outline the synthesis of reserpine.	5
2	224.	Write a note on the synthesis of reserpine.	5
2	225.	How do you elucidate the structure of lysergic acid?	5
2	226.	Elucidate the structure of lysergic acid.	5
2	227.	Explain the elucidation of the structure of lysergic acid.	5
2	228.	Outline the structure of lysergic acid.	5
2	229.	Elaborate the structure of lysergic acid.	5
2	230.	Give the synthesis of lysergic acid.	5
2	231.	Explain the synthesis of lysergic acid.	5
2	232.	Describe the synthesis of lysergic acid.	5
2	233.	Outline the synthesis of lysergic acid.	5
2	234.	Write a note on the synthesis of lysergic acid.	5
2	235.	How do you elucidate the structure of nicotine?	5
2	236.	Elucidate the structure of nicotine.	5
2	237.	Explain the elucidation of the structure of nicotine.	5
2	238.	Outline the structure of nicotine.	5
2	239.	Elaborate the structure of nicotine.	5
2	240.	Give the biosynthesis of nicotine.	5
2	241.	Write a note on biosynthesis of nicotine.	5
2	242.	Explain the biosynthesis of nicotine.	5
2	243.	Outline the biosynthesis of nicotine.	5
2	244.	Describe the biosynthesis of nicotine.	5
2	245.	Elaborate the biosynthesis of nicotine.	5
2	246.	Give the biosynthesis of morphine.	5
2	247.	Write a note on biosynthesis of morphine.	5

2	248.	Explain the biosynthesis of morphine.	5
2	249.	Outline the biosynthesis of morphine.	5
2	250.	Describe the biosynthesis of morphine.	5
2	251.	Elaborate the biosynthesis of morphine.	5
3	252.	Write the structural elucidation of vitamin-A_1	5
3	253.	Write the synthesis of vitamin-A_1	5
3	254.	Discuss the Werner-Milar method for the synthesis of vitamin-A_1	5
3	255.	Write the synthesis of vitamin-A_1 by Van-dorp method.	5
3	256.	Write the synthesis of vitamin-A_1 by Pommer method	5
3	257.	Discuss the structural elucidation of vitamin-B_1	5
3	258.	Write the synthesis of thiamine.	5
3	259.	Discuss the synthesis of thiamine by Williams method.	5
3	260.	Write the constitution of vitamin-B_6	5
3	261.	Write the synthesis of pyridoxine	5
3	262.	Discuss the constitution of Biotin.	5
3	263.	How do you synthesize biotin from 2-thienyl valericacid.	5
3	264.	Elucidate the structure of ascorbic acid	5
3	265.	Explain the biosynthesis of ascorbic acid	5
3	266.	Write the synthesis of vitamin-C	5
3	267.	Elucidate the structure of vitamin-E	5
3	268.	Write the synthesis of vitamin –E	5
3	269.	Discuss the structural elucidation of vitamin-K	5
3	270.	Elucidate the structure of vitamin-A_1	5
3	271.	Elucidate the structure of vitamin-H	5
3	272.	Write the structure of vitamin-K. Mention its importance	5

3	273.	Give a brief account of biosynthesis of Vitamin-C	5
3	274.	Outline the synthesis of Riboflavin	5
3	275.	How do you synthesis thiamine.	5
3	276.	Discuss the synthesis of vitamin-A and explain about conversion of vitamin $A_1$to A_2	5
3	277.	Explain isoalloxazine ring system is present in riboflavin.	5
3	278.	Explain 1,4- naphthaquinone ring in Vitamin-K	5
3	279.	Elucidate the structure of Vitamin-E	5
3	280.	Discuss the synthesis of ascorbic acid	5
3	281.	Elucidate the structure of riboflavin	5
3	282.	Discuss the constitution of thiamine	5
3	283.	Write the structural elucidation of Vitamin-H	5
3	284.	Discuss the constitution of Vitamin-H	5
3	285.	Discuss the constitution of biotin	5
3	286.	Write the structural elucidation of biotin	5
3	287.	Write the Almgnist synthesis of Vitamin-K_1	5
3	288.	Discuss the William's method for the synthesis of thiamine.	5
3	289.	Elucidate the structure of pyridoxine.	5
3	290.	Write the synthesize of Vitamin B_6 by using cyanoacetamide.	5
3	291.	Discuss the constitution of phylloquinone.	5
3	292.	How do you synthesize biotin from 2-thienyl valericacid.	5
3	293.	Write the constitution of retinol	5
3	294.	Discuss the structural elucidation of 3,4dehydroretinol.	5
2	295.	Give the classification and isolation of alkaloids.	6

2	296.	Outline the classification and isolation of alkaloids.	6
2	297.	Elaborate the classification and isolation of alkaloids.	6
2	298.	Write a note on the classification and isolation of alkaloids.	6
2	299.	Explain the general methods for determining the structures of alkaloids.	6
2	300.	Describe the general methods for determining the structures of alkaloids.	6
2	301.	Outline the methods for determining the structures of alkaloids.	6
2	302.	Elaborate the general methods for determining the structures of alkaloids.	6
2	303.	Write a note on general methods for determining the structures of alkaloids.	6
2	304.	How do you elucidate the structure of quinine?	6
2	305.	Elucidate the structure of quinine.	6
2	306.	Explain the elucidation of the structure of quinine.	6
2	307.	Outline the structure of quinine.	6
2	308.	Elaborate the structure of quinine.	6
2	309.	Give the synthesis of quinine.	6
2	310.	Explain the synthesis of quinine.	6
2	311.	Describe the synthesis of quinine.	6
2	312.	Outline the synthesis of quinine.	6
2	313.	Write a note on the synthesis of quinine.	6
2	314.	How do you elucidate the structure of morphine?	6
2	315.	Elucidate the structure of morphine.	6
2	316.	Explain the elucidation of the structure of morphine.	6
2	317.	Outline the structure of morphine.	6
2	318.	Elaborate the structure of morphine.	6
2	319.	Give the synthesis of morphine.	6
2	320.	Explain the synthesis of morphine.	6

2	321.	Describe the synthesis of morphine.	6
2	322.	Outline the synthesis of morphine.	6
2	323.	Write a note on the synthesis of morphine.	6
2	324.	How do you elucidate the structure of reserpine?	6
2	325.	Elucidate the structure of reserpine.	6
2	326.	Explain the elucidation of the structure of reserpine.	6
2	327.	Outline the structure of reserpine.	6
2	328.	Elaborate the structure of reserpine.	6
2	329.	Give the synthesis of reserpine.	6
2	330.	Explain the synthesis of reserpine.	6
2	331.	Describe the synthesis of reserpine.	6
2	332.	Outline the synthesis of reserpine.	6
2	333.	Write a note on the synthesis of reserpine.	6
2	334.	How do you elucidate the structure of lysergic acid? Give its applications	6
2	335.	Elucidate the structure and applications of lysergic acid.	6
2	336.	Explain the elucidation of the structure of lysergic acid. Give its applications.	6
2	337.	Outline the structure and applications of lysergic acid.	6
2	338.	Elaborate the structure and applications of lysergic acid.	6
2	339.	Give the synthesis and applications and applications of lysergic acid.	6
2	340.	Explain the synthesis and applications of lysergic acid.	6
2	341.	Describe the synthesis and applications of lysergic acid.	6
2	342.	Outline the synthesis and applications of lysergic acid.	6
2	343.	Write a note on the synthesis and applications of lysergic acid.	6
2	344.	Give the biosynthesis of nicotine.	6
2	345.	Write a note on biosynthesis of nicotine.	6

2	346.	Explain the biosynthesis of nicotine.	6
2	347.	Outline the biosynthesis of nicotine.	6
2	348.	Describe the biosynthesis of nicotine.	6
2	349.	Elaborate the biosynthesis of nicotine.	6
2	350.	Give the biosynthesis of morphine.	6
2	351.	Write a note on biosynthesis of morphine.	6
2	352.	Explain the biosynthesis of morphine.	6
2	353.	Outline the biosynthesis of morphine.	6
2	354.	Describe the biosynthesis of morphine.	6
2	355.	Elaborate the biosynthesis of morphine.	6

Question Paper Pattern- Model Question Paper

4.	a.	Elucidate the structure of cholesterol	07
	b.	Write a note on classification of alkaloids.	03
5.	a.	Discuss the synthesis of vitamin-A and explain about conversion of vitamin A_1 to A_2	10
6.	a.	Elucidate the structure of Reserpine	06
	b.	Write the structure of Lysergic acid and mention its applications	04
7.	a.	Give the biosynthesis of nicotine	06
	b.	How do you synthesis thiamine?	04
8.	a.	Outline the synthesis of Riboflavin	04
	b.	How do you elucidate the structure of morphine?	06

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